

PFAS exposure in Belgium: safe for now, but are we looking at the whole picture?

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The latest EFSA evaluation on per- and polyfluoroalkyl substances (PFAS) resulted in a tolerable weekly intake (TWI) for the sum of PFOS, PFOA, PFHxS and PFNA ($\Sigma 4$ PFAS) of 4.4 ng/kg bw/week [1]. Given that the last Belgian exposure assessment focused solely on PFOS and PFOA [2], a re-evaluation of the exposure and risk to PFAS was needed.

A comprehensive sampling, reflecting all foods relevant for PFAS exposure and Belgian consumption habits, yielded 283 food samples. The study did not include homegrown food. The selection of 25 PFAS was based on Recommendation (EU) 2022/1431 and Regulation (EU) 2023/915. The samples were analysed using a validated method, involving a QuEChERS-based extraction, a two-step SPE purification process, and analysis by UHPLC-HRMS. Potential PFAS contaminations in the laboratory were minimised to achieve low quantification limits. The PFAS occurrence data [3] were combined with consumption data from the Belgian food consumption survey [4] to assess dietary exposure. The risk assessment for $\Sigma 4$ PFAS was conducted following the methodology used by EFSA [1].

Of the 25 targeted PFAS, a quantifiable exposure assessment could be conducted for 13. The exposure estimates were highest for PFBA, followed by a PFAS group with exposure estimates approximately ten times lower: PFPeA, PFOA, PFHxA, and PFOS. The exposure to the remaining PFAS was at least 30-fold lower. The middle-bound dietary exposure to PFOS in Belgium decreased 8-fold compared to the previous Belgian study from 2012, which may reflect the phase-out of PFOS in the EU since 2009. The exposure to PFOA, phased out since 2020, also decreased, but to a lesser extent.

The mean exposure to the $\Sigma 4$ PFAS, assuming equal toxicity potencies, ranged from 0.93 ng/kg bw/week for adults to 1.7 ng/kg bw/week for children. The TWI was not exceeded for adolescent and adult populations and was only marginally exceeded by a minor fraction of the children's population. Hence, if solely dietary exposure to $\Sigma 4$ PFAS is considered, no health concerns are expected for the Belgian population. Nonetheless, exposure to PFAS extends beyond these four compounds and is not restricted to food alone.

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[1] EFSA. EFSA Journal 18, e06223 (2020)

[2] Bervoets et al. Eindrapport RF1681. (2012)

[3] Van Leeuw et al. Chemosphere 366, 143543 (2024)

[4] Bel et al. Arch Public Health 74, 20 (2016)